J.17

NAME ☞	
SCHOOL ☞	
TEACHER ☞	

Pre-Junior Certificate Examination, 2017

Mathematics

Paper 1
Ordinary Level

Time: 2 hours

300 marks

For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

School stamp

Running total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if you do not show all necessary work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

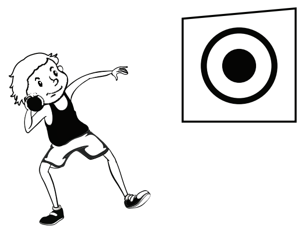
Question 1 **(Suggested maximum time: 10 minutes)**

(a) Fill in the missing numbers in the boxes below so that the calculations are correct.

(i) $96 + 28 \quad = \quad 73 + \boxed{}$

(ii) $36 \times 4 \quad = \quad \boxed{} \times 6$

(b) Tim plays a game of throwing balls at a target.



He throws six balls at the target, one at a time.

Each hit is worth 5 points.
Each miss is worth -3 points.

Tim hits the target with three of the balls and misses with the rest.

(i) How many points does Tim score? Answer = $\boxed{}$

Lara throws seven balls at the target, one at a time.

(ii) Is it possible for Lara to score the same number of points as Tim?
Give a reason for your answer.

Answer:

Reason:

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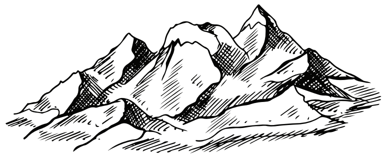
Question 2 **(Suggested maximum time: 5 minutes)**

(a) The heights of five mountains in Ireland are listed below.

Mountain	Height (in metres)
Galtymore	919
Carrauntoohil	1038
Errigal	751
Lugnaquilla	925
Croagh Patrick	764



(i) Arrange the mountains in order of height, starting with the highest.

Highest: ______________________

____

____

____

Lowest: ______________________

(ii) Find the difference in height between the highest mountain and the lowest mountain.

(iii) The highest mountain in Europe is Mount Elbrus. Its height is 5642 metres.

Write down its height to the nearest 1000 metres. Answer = ☐

(b) Write the number 875·46

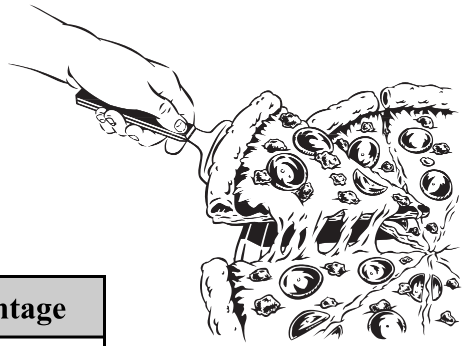
(i) correct to the nearest **whole number**. ☐

(ii) correct to the nearest **100**. ☐

(iii) correct to **2 significant figures**. ☐

(iv) correct to **1 decimal place**. ☐

Question 3 **(Suggested maximum time: 10 minutes)**



Sam, Lauren and Chris share a pizza.

Sam ate 40% of the pizza, Lauren ate $\frac{1}{4}$ of the pizza and Chris ate 0·3 of the pizza.

(a) The numbers in the table below are shown in fraction, decimal and percentage form. Complete the table.

	Fraction	**Decimal**	**Percentage**
Sam	$\frac{2}{5}$		40%
Lauren	$\frac{1}{4}$	0·25	
Chris		0·3	

(b) Which person ate the largest amount of the pizza? Answer =

Give a reason for your answer.

Reason:

(c) How much of the pizza was left over?

(d) The price of the pizza was €13·95. There was also a €4·50 delivery charge.
Sam, Lauren and Chris shared the total cost of the pizza equally.
How much did each pay?

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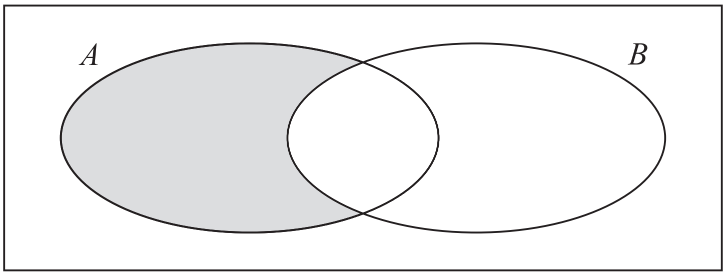
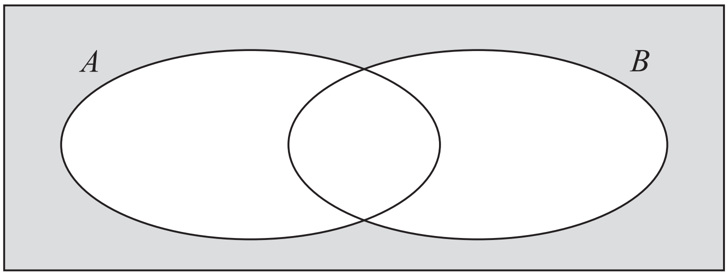
Question 4 **(Suggested maximum time: 10 minutes)**

(a) Describe in your own words what the shaded regions in each of the following Venn diagrams represent.

(i)



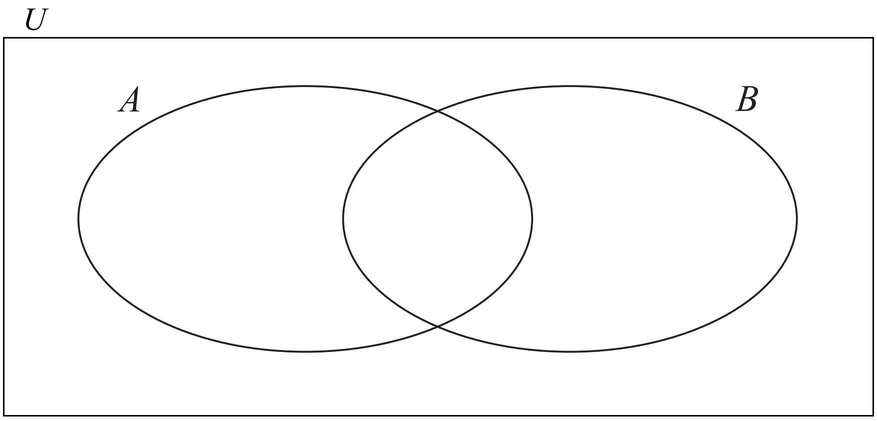
(ii)



(b) The sets U, A, and B are defined as follows:

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
A is the set of even numbers, from 1 to 10 inclusive.
B is the set of multiples of 3, from 1 to 10 inclusive.

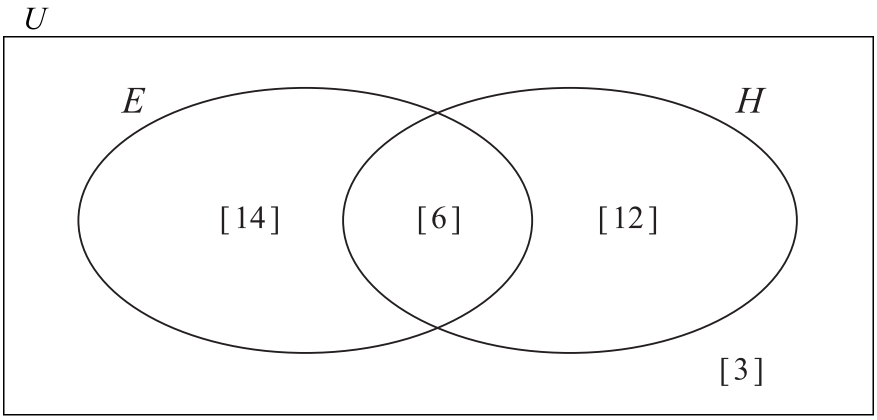
(i) Use this information to fill in the Venn diagram below.



(ii) Write down $\#A$. Answer =

(iii) Explain why $A \cup B \neq U$.

(c) The number of students in a class who use an e-reader (E) and the number of students in a class who use 'hardcopy' books (H) are shown in the Venn diagram below.



(i) How many students use an e-reader?

(ii) How many students use an e-reader and 'hardcopy' books?

(iii) How many students read 'hardcopy' books only?

(iv) How many students are in the class?

(v) Suggest a reason why three students are not in Set E or Set H.

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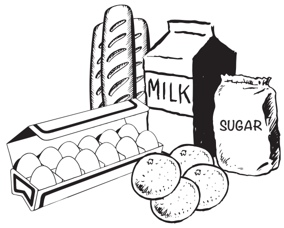
Question 5 **(Suggested maximum time: 10 minutes)**

Paul estimates the total cost of his groceries while he is shopping.



To estimate the total cost, he rounds the cost of each item to the nearest euro and then adds the estimates together.

(a) Fill in the table below to show Paul's calculations.
One has already been filled in for you.

Item	Price per unit	Estimate
2 kg of sugar	€1·39 per kg	2 × €1·00 = €2·00
3 litres of milk	€1·18 per litre	× =
6 bread rolls	€0·79 per roll	× =
1 dozen eggs	€3·25 per dozen	× =
4 oranges	€0·55 each	× =
Total Estimate		

(b) Fill in the table below to calculate the actual cost of Paul's groceries.

Item	Price per unit	Actual Cost
2 kg of sugar	€1·39 per kg	
3 litres of milk	€1·18 per litre	
6 bread rolls	€0·79 per roll	
1 dozen eggs	€3·25 per dozen	
4 oranges	€0·55 each	
Total Cost		

(c) Do you think Paul's estimate was near the actual cost? Yes ☐ No ☐

Give a reason for your answer.

Reason:

(d) Paul has a €50 note. He can only spend 25% of this money on groceries. Will he have enough money to buy all the items? Yes ☐ No ☐

Give a reason for your answer.

Reason:

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Question 6 **(Suggested maximum time: 10 minutes)**

Colin is making some pancakes from the recipe shown.

This recipe makes **12** pancakes.

Ingredients
120 g of flour
2 eggs
250 ml of milk

(a) Colin wishes to make **30** pancakes.
Find the amount of each ingredient needed for 30 pancakes.
Write the amounts needed in the boxes to the right.

Ingredients	
☐	g of flour
☐	eggs
☐	ml of milk

(b) Colin sells all the pancakes he makes to a local café.
It costs him €8 to make the pancakes and he sells them for €14.

(i) Express the profit Colin made as a percentage of the cost price.

(ii) The café makes 88% profit on the pancakes.
How much does the café make on the pancakes?

Question 7 **(Suggested maximum time: 5 minutes)**

The table below shows the values when 4 is raised to certain powers.

(a) Complete the table.

Power of 4	Expanded power of 4	Answer
4^1	4	4
4^2	4×4	16
	$4 \times 4 \times 4$	64
4^4	$4 \times 4 \times 4 \times 4$	
4^5		1024
4^6	$4 \times 4 \times 4 \times 4 \times 4 \times 4$	
		16·384
4^8		65·536

(b) Use the table to help you write $1024 \times 16 = 16\,384$ in the form 4^p, where $p \in \mathbb{N}$.

(c) Use the table to help you work out the value of $\frac{65\,536}{1024}$ in the form 4^p, where $p \in \mathbb{N}$.

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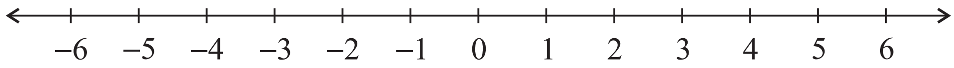
Question 8 **(Suggested maximum time: 10 minutes)**

(a) Write down **one** value of x which satisfies the inequality $7x > 42$, where $x \in \mathbb{N}$.

Answer =

(b) **(i)** Solve the inequality $2x - 1 < 7$, where $x \in \mathbb{N}$.

(ii) Graph your solution to part **(i)** on the number line below.



(c) **(i)** Express $\frac{x}{2} + \frac{3x+1}{5}$ as a single fraction. Give your answer in its simplest form.

$\frac{x}{2} + \frac{3x+1}{5} =$

(ii) Using your answer from part **(i)**, or otherwise, solve the equation $\frac{x}{2} + \frac{3x+1}{5} = 9$.

Question 9 **(Suggested maximum time: 10 minutes)**

(a) $g(x) = 5x - 2$. Find

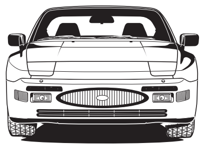
(i) $g(3)$

$g(3) =$ ☐

(ii) $g(-1)$

$g(-1) =$ ☐

(b) Ann decides to hire a car during her holidays. The cost of hiring a car for a number of days is calculated using the formula:



Hire Cost = €25 × Number of days + €100.

(i) How much will it cost Ann if she decides to hire a car for 7 days?

(ii) Ann hires a car. The hire cost is €250.
How many days does she hire the car for?

(iii) Sarah has €440 to spend on hiring a car.
What is the **maximum** number of days she can hire a car for?

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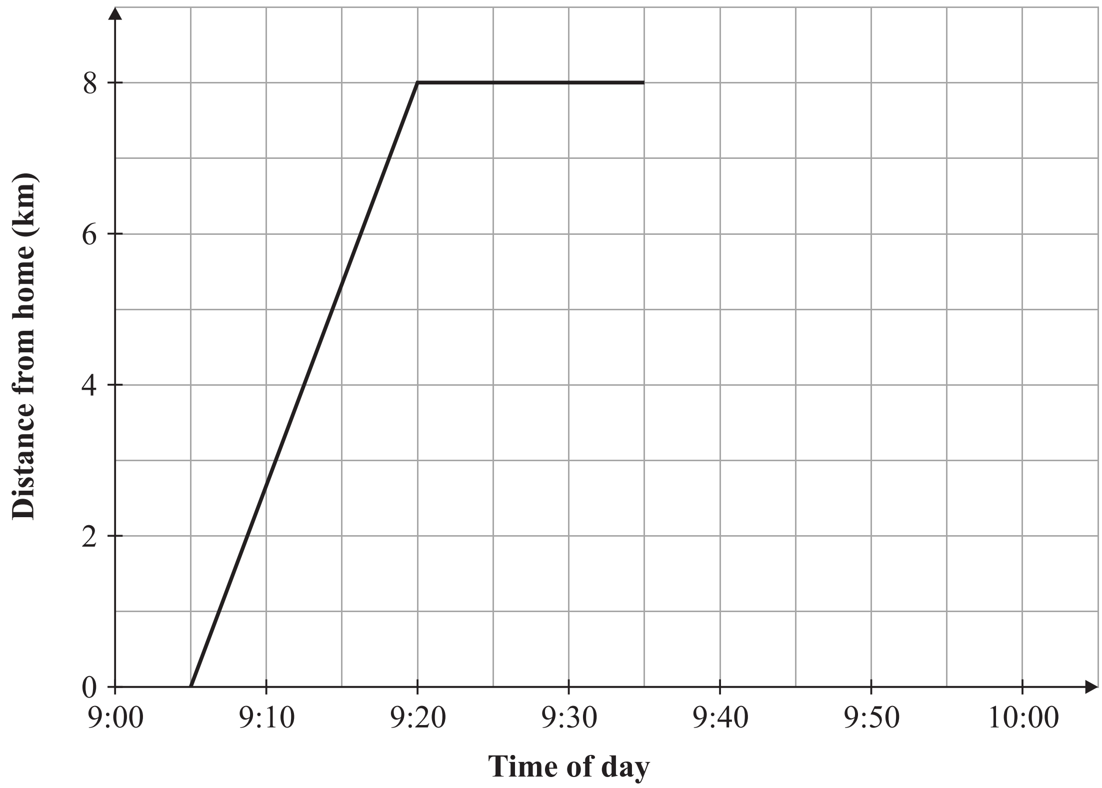
Question 10 **(Suggested maximum time: 10 minutes)**

Debbie cycled from her home to her friend's house.
She stayed a while at her friend's house and then travelled home again.
The graph represents part of her journey.



(a) At what time did Debbie leave home? Answer = ☐

(b) How far was Debbie from home while visiting her friend? Answer = ☐ km

(c) How long did Debbie stay at her friend's house? Answer = ☐ minutes

Debbie arrived home at 9:55.

(d) Use this information to complete the graph above.

(e) Debbie says "Cycling home was quicker than cycling to my friend's house."

Is Debbie correct? Yes ☐ No ☐

Give a reason for your answer.

Reason:

Question 11 **(Suggested maximum time: 10 minutes)**

(a) Factorise fully each of the following.

(i) $8x + 24$

(ii) $x^2 - 100$

(iii) $x^2 - 13x + 30$

(b) Two expressions are written on the cards below.

$\boxed{5x - 7}$ $\boxed{2x + 5}$

What value of x makes the cards equal?

(c) **(i)** Simplify $2(x + 4y) + 6(3x + y)$.

(ii) Simplify $3x(2x^2 - 1)$.

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Question 12 **(Suggested maximum time: 10 minutes)**

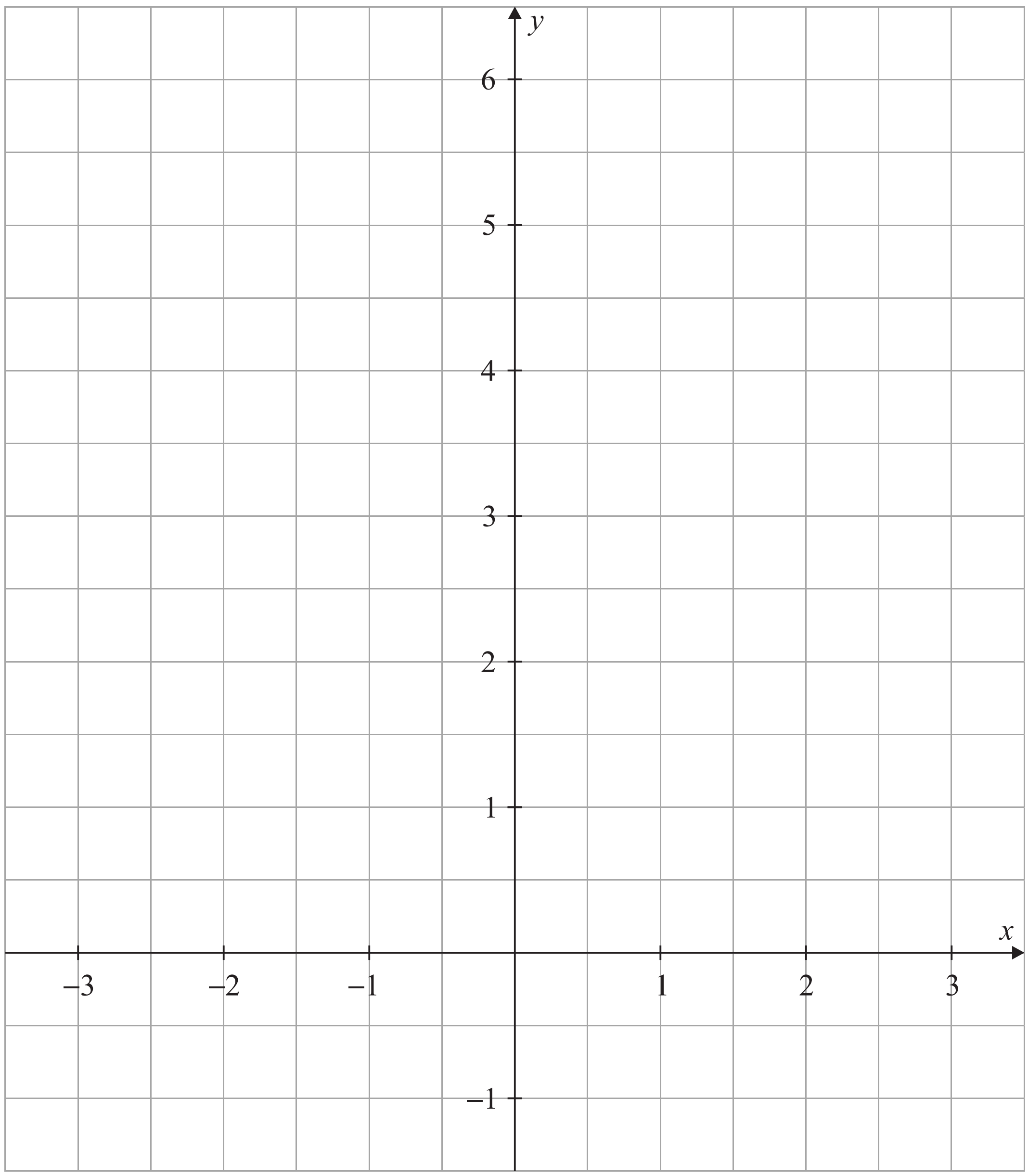
(a) Complete the table for the function

$f : x \mapsto x^2 + x$

in the domain $-3 \le x \le 2$, where $x \in \mathbb{R}$.

x	$f(x)$
-3	6
-2	
-1	0
0	
1	
2	

(b) Using the values in the table, draw the graph of the function $f : x \mapsto x^2 + x$ in the domain $-3 \le x \le 2$, where $x \in \mathbb{R}$.



The function $f : x \mapsto x^2 + x$ gives the predicted average monthly air temperature, in degrees Celsius (°C), over a 5-month period of time in Copenhagen.

The x-axis represents the months: November ($x = -3$), December ($x = -2$), January ($x = -1$), February ($x = 0$), March ($x = 1$) and April ($x = 2$).

Use your graph from part **(b)** to answer the following questions. Show your work on the graph.

(c) What is the predicted average monthly air temperature for January?

Answer = ☐ °C

(d) Find the months when the predicted average monthly air temperature is 2 °C.

Answer = ☐ and ☐

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Pre-Junior Certificate, 2017 – Ordinary Level

Mathematics – Paper 1

Time: 2 hours